

Project Initialization and Planning Phase

Date	26 June 2026
Team ID	
Project Name	Smart Lender
Maximum Marks	5 Marks

Project Proposal (Proposed Solution) report

The proposed solution aims to simplify and accelerate the loan approval process using Artificial Intelligence and Machine Learning. The system predicts whether a loan applicant is eligible based on financial and personal information. This helps financial institutions reduce manual effort, improve prediction accuracy, minimize risks, and provide faster loan decisions to customers.

Project Overview	
Objective	The primary objective is to develop an AI-powered loan eligibility prediction system that accurately predicts whether an applicant qualifies for a loan, enabling faster, fairer, and more efficient decision-making.
Scope	The project focuses on collecting applicant details, preprocessing data, training a machine learning model, predicting loan eligibility, and displaying instant results through a user-friendly web application.
Problem Statement	
Description	Traditional loan approval processes are time-consuming, involve extensive manual verification, and may result in inconsistent decisions. Customers often experience delays and uncertainty while waiting for approval.
Impact	Implementing an AI-based prediction system reduces processing time, improves decision accuracy, minimizes operational costs, and enhances customer satisfaction.
Proposed Solution	
Approach	The system uses Machine Learning algorithms to analyze applicant information such as income, loan amount, credit history, employment status, and other financial factors to predict loan approval.
Key Features	• AI-based loan eligibility prediction • Fast and accurate loan approval decisions • User-friendly web interface • Reduced manual verification • Secure handling of applicant information • Improved customer experience

Resource Requirements

Resource Type	Description	Specification/ Allocation
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Hardware		
Computing Resources	Processor configuration for development and model training	Intel i5/Ryzen 5 or higher
Memory	RAM for smooth development and execution	Minimum 8 GB RAM
Storage	Disk space for datasets, models, application and logs	256 GB SSD or above
Software		
Programming Language	Language used for application and model	Python 3.10+
Framework	Web framework for building application	Flask
Libraries	ML and data processing libraries	Scikit-learn, Pandas, NumPy, Matplotlib
Development Environment	IDE for development and testing	Visual Studio Code / PyCharm / Jupyter Notebook
Database		
Database Management System	For storing user data and prediction history	MySQL
Data		
Dataset	Source, size and format of dataset	Loan Prediction Dataset (CSV) from Kaggle